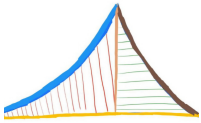
Math for EE



G. V. V. Sharma

About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. CBSE-2020

A. Formulae-10TH

1.1.1

$$b^2 - 4ac = 0 \tag{1.1.1.1}$$

1.1.2

$$V = \frac{4}{3}\pi r^3 \tag{1.1.2.1}$$

1.1.3 $\mathbf{x} = \begin{bmatrix} x_2 - x_1 \\ y_2 - y_1 \end{bmatrix}$, d is the distance between points

$$d = \sqrt{\mathbf{x}^T\mathbf{x}} \tag{1.1.3.1}$$

1.1.4

$$P(x) = Q(x)Y(x) + R(x) \tag{1.1.4.1}$$

1.1.5 from Basic proportionality theorem

$$\frac{AD}{DB} = \frac{AE}{EC} \tag{1.1.5.1}$$

1.1.6 equation of line

$$\frac{x}{a} + \frac{y}{b} = 1 \tag{1.1.6.1}$$

B. Problems-10TH

1.2.1 The value(s) of k for which the quadratic equation $2x^2 + kx + 2 = 0$ has equal roots, is

a) 4 b) ± 4 c) -4 d) 0

Solution:
For a quadratic equation $ax^2 + bx + c = 0$,condition to have equal roots from (1.1.1.1)

$$b^2 - 4ac = 0 \tag{1.2.1.1}$$

$k^2 - 16 = 0 \implies k = \pm 4$

codes/10/q1.py (1.2.1.2)

1.2.2 which of the following is not an A.P?

a) -1.2,0.8,2.8 ...
b) $3, 3+\sqrt{2}, 3+2\sqrt{2}, ...$
c) $\frac{4}{3}, \frac{7}{3}, \frac{9}{3}, \frac{12}{3}...$
d) $\frac{-1}{5}, \frac{-2}{5}, \frac{-3}{5}....$

Solution:

a.$a_2 - a_1 = 2, a_3 - a_2 = 2$, so,they form an A.P with common difference 2

b.$a_2 - a_1 = \sqrt{2}, a_3 - a_2 = \sqrt{2}$, they form an A.P with common difference $\sqrt{2}$

c.$a_3 - a_2 = \frac{2}{3}, a_4 - a_3 = 1$, no common difference ,so they doesn't form an A.P

1.2.3 The radius of a sphere (in cm) whose volume is 12π cm^3 is

a) 3 b) $3\sqrt{3}$ c) $3^{2/3}$ d) $3^{1/3}$

Solution:

from (1.1.2.1)

volume of the sphere

$$V = \frac{4}{3}\pi r^3 cm^3 \tag{1.2.3.1}$$

$\frac{4}{3}\pi r^3 = 12\pi cm^3 \implies r = 3^{\frac{2}{3}} cm$

1.2.4 The distance between the points $(m, -n)$ and $(-m, n)$ is

a) $\sqrt{m^2 + n^2}$ b) $m + n$ c) $2\sqrt{m^2 + n^2}$ d) $\sqrt{2m^2 + 2n^2}$

Solution:

from (1.1.3.1)

distance between two points $(x_1, y_1); (x_2, y_2)$ is

$$d = \sqrt{\mathbf{x}^T\mathbf{x}} \tag{1.2.4.1}$$

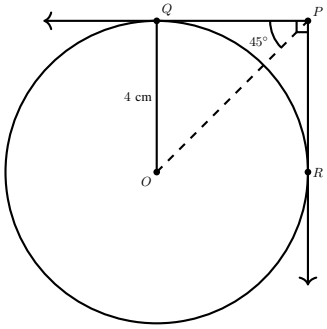
$\mathbf{x} = \begin{bmatrix} 2m \\ -2n \end{bmatrix}$

distance $d = \sqrt{4m^2 + 4n^2} \implies 2\sqrt{m^2 + n^2}$

codes/q4.c (1.2.4.2)

1.2.5 In Figure-1, from an external point P, two tangents PQ and PR are drawn to a circle of radius 4 cm with centre O. If $\angle QPR = 90°$, then length of PQ is

a) 3 cm b) 4 cm c) 2 cm d) $2\sqrt{2}$cm



Solution: The line joining the centre to the point of intersection bisects the angle between the tangents.
so $\angle OPQ = \frac{\pi}{4}$ $tan(\frac{\pi}{4}) = \frac{OP}{PQ}$
$PQ = OP \implies PQ = 4cm$

1.2.6 On dividing a polynomial $p(x)$ by $x^2 - 4$, quotient and remainder are found to be x and 3 respectively. The polynomial $p(x)$ is

a) $3x^2 + x - 12$ b) $x^3 - 4x + 3$ c) $x^2 + 3x - 4$ d) $x^3 - 4x - 3$

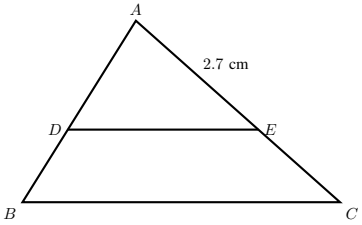
Solution: from (1.1.4.1)

$$P(x) = Q(x)Y(x) + R(x) \tag{1.2.6.1}$$

$P(x), Y(x) = x^2 - 4, Q(x) = x, R(x) = 3$
$P(x) = x^3 - 4x + 3$

1.2.7 In Figure-2, $DE \parallel BC$. If $\frac{AD}{DB} = \frac{3}{2}$ and $AE = 2 \cdot 7$ cm, then EC is equal to

a) $2 \cdot 0$ cm b) $1 \cdot 8$ cm c) $4 \cdot 0$ cm d) $2 \cdot 7$ cm



Solution:
Basic Proportionality theorem : If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then it divides those two sides in the same ratio.
form (1.1.5.1)
According to the theorem , $\frac{AD}{DB} = \frac{AE}{EC}$
$\frac{2}{3} = \frac{EC}{AE} \implies EC = 1.8cm$

1.2.8 The point on the x-axis which is equidistant from $(-4, 0)$ and $(10, 0)$ is

a) $(7, 0)$ b) $(5, 0)$ c) $(0, 0)$ d) $(3, 0)$

Solution:
let ,the point be P(x,0), Q(10,0),R(-4,0) . Distance between $|PQ| = |PR|$
$|x - 10| = |x + 4| \implies x = 3$

1.2.9 The centre of a circle whose end points of a diameter are $(-6, 3)$ and $(6, 4)$ is

a) $(8,-1)$ b) $(4,7)$ c) $\left(0,\frac{7}{2}\right)$ d) $\left(4,\frac{7}{2}\right)$

Solution:
centre $\implies$ midpoint of end points of diameter
centre $O(x,y)=(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}) \implies O(x,y)=(0,\frac{7}{2})$

1.2.10 The pair of linear equations $\frac{3x}{2}+\frac{5y}{3}=7$ and $9x+10y=14$ is

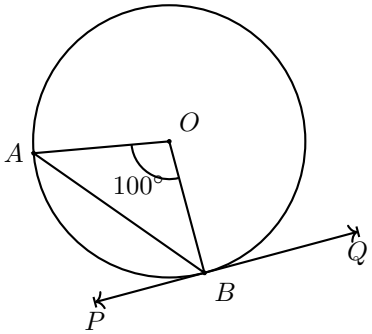
a) consistent
b) inconsistent
c) consistent with one solution
d) consistent with many solutions

Solution:
$9x+10y=42$ and $9x+10y=14$, they are parallel lines , so lines are inconsistent

codes/10/q9.py (1.2.10.1)

1.2.11 In Figure-3, PQ is tangent to the circle with centre at O, at the point B. If $\angle AOB=100^\circ$, then $\angle ABP$ is equal to

a) 50° b) 40° c) 60° d) 80°



Solution:
$\angle OAB=\angle OBA=40^\circ$
$\angle OBP=90^\circ \implies \angle ABP=50^\circ$

1.2.12 Simplest form of $\frac{1+\tan^2 A}{1+\cot^2 A}$ is

Solution:

$$\frac{1+\frac{sin^2A}{cos^2A}}{1+\frac{cos^2A}{sin^2A}} \implies \frac{sin^2A}{cos^2A}$$

$$=tan^2A$$

1.2.13 If the probability of an event E happening is $0\cdot023$, then $P(\overline{E})=$

solution :

$$P(E)+P(\overline{E})=1$$

$$P(\overline{E}) = 1 - P(E) \implies P(\overline{E}) = 0.977$$

1.2.14 A cone and a cylinder have the same radii but the height of the cone is 3 times that of the cylinder. Find the ratio of their volumes.

Solution:

let V_1 and V_2 be the volumes of cone and cylinder

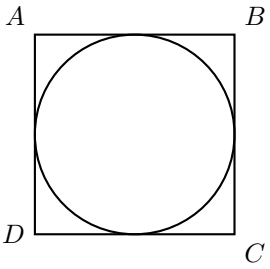
$r_1 = r_2; h_1 = 3h_2$

$$V_1 = \frac{1}{3}\pi r_1^2 h_1$$

$$V_2 = \pi r_2^2 h_2 \implies \frac{V_1}{V_2} = \frac{h_1}{3h_2} = 1$$

$$\frac{V_1}{V_2} = 1$$

1.2.15 In Figure-4, a quadrilateral ABCD is drawn to circumscribe a circle. Prove that $AB + CD = BC + AD$



Solution:

let the point of intersections of quadrilateral and circle be P,Q,R,S

side AB at P, side BC at Q, side CD at R, side AD at S

$\overline{AP} = \overline{AS}$

$\overline{BP} = \overline{BQ}$,

$\overline{CQ} = \overline{CR}$,

$\overline{DR} = \overline{DS}$

$AP + PB = AB; BQ + QC = BC; CR + RD = CD; DS + AS = AD$

$$AB + CD = AP + PB + CR + RD$$

$$AD + BC = AS + BQ + CQ + DS \implies AP + PB + CR + RD$$

$$AB + CD = AD + BC$$

Hence proved .

1.2.16 Find the sum : (-5)+(-8)+(-11)+ ...+(-230)

Solution: Find the sum of the series:

$$(-5) + (-8) + (-11) + \cdots + (-230)$$

First term (a) = -5
Common difference (d) = $(-8) - (-5) = -3$
Last term (l or a_n) = -230
Using the formula for the n-th term of an A.P., $a_n = a + (n-1)d$:

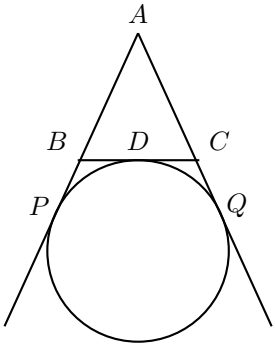
$$\begin{aligned} -230 &= -5 + (n-1)(-3) \\ -230 + 5 &= -3(n-1) \\ -225 &= -3(n-1) \\ n - 1 &= \frac{-225}{-3} \\ n - 1 &= 75 \\ n &= 76 \end{aligned}$$

Using the formula for the sum of an A.P., $S_n = \frac{n}{2}(a + l)$:

$$\begin{aligned} S_{76} &= \frac{76}{2}\left[(-5) + (-230)\right] \\ S_{76} &= 38 \times (-235) \\ S_{76} &= -8930 \end{aligned}$$

The sum of the given A.P. series is -8930

1.2.17 In Figure-5, find the perimeter of ΔABC if AP = 12 cm



Solution:
let $BP = BD = CD = CQ = xcm$ $AP = AQ = 12cm$
$AB = AC = 12 - xcm$
perimeter of $\triangle ABC = 2x + 12 - x + 12 - x$
perimeter = 24cm

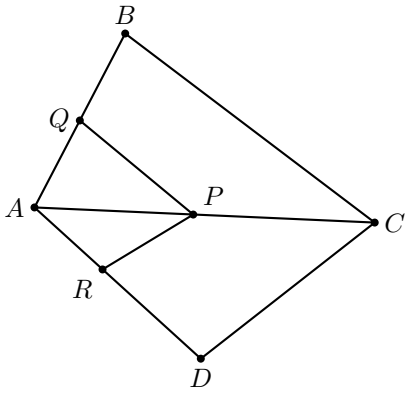
1.2.18 Find the mode of the following distribution :

marks	0-10	10-20	20-30	30-40	40-50	50-60
number of students	4	6	7	12	5	6

Solution:
modal class : 30-40
l=30
modal frequency (f_1): 12

$f_0 = 7$
$f_2 = 5$
class size h= 10
mode = $l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) h \implies mode = 30 + \frac{25}{6}$
mode = 34.166

1.2.19 In Figure-7, if $PQ \parallel BC$ and $PR \parallel CD$, prove that $\frac{QB}{AQ} = \frac{DR}{AR}$



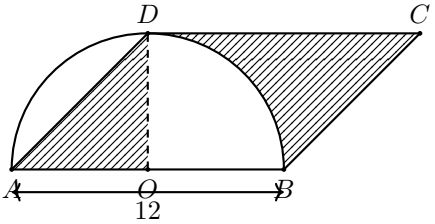
solution :
For the $\triangle AQP and \triangle ABC$, given $QP||BC$, by Thale's theorem $\frac{AQ}{QB} = \frac{AP}{PC}$
for the $\triangle ACD and \triangle APR$, given $RP||CD$, by Thale's theorem $\frac{AR}{RD} = \frac{AP}{PC}$

$$\frac{AQ}{QB} = \frac{AR}{RD}$$

1.2.20 Prove that: $(\sin^4\theta - \cos^4\theta + 1)cosec^2\theta = 2$
solution :

$$((sin^2\theta - cos^2\theta)(cos^2\theta + sin^2\theta) + 1)cosec^2\theta$$
$$(1 - cos(2\theta))cosec^2\theta \implies 2sin^2\theta cosec^2\theta = 2$$

1.2.21 In Figure-8, ABCD is a parallelogram. A semicircle with centre O and the diameter AB has been drawn and it passes through D. If $AB = 12$ cm and $OD \perp AB$, then find the area of the shaded region. (Use $\pi = 3 \cdot 14$)



Solution:
$AB = CD = 12cm$ and $AD = BC = 6\sqrt{2}cm$

area of shaded region = area of parallelogram - area of quarter circle
$r = 6cm$, area of quarter circle $= \frac{1}{4}\pi r^2 = 9\pi cm^2$
area of parallelogram $= b.h$; b= length of base
h= perpendicular length from base to opposite side
area of paralellogram = $12 * 6 = 72cm^2$
area of shaded region $= 72 - 9\pi cm^2 = 43.725cm^2$

1.2.22 A fraction becomes $\frac{1}{3}$ when 1 is subtracted from the numerator and it becomes $\frac{1}{4}$ when 8 is added to its denominator. Find the fraction.

Solution:

let fraction be $\frac{x}{y} \implies \frac{x-1}{y} = \frac{1}{3}$

$$3x - y = 4 \tag{1.2.22.1}$$

$\frac{x}{y+8} = \frac{1}{4}$

$$4x - y = 8 \tag{1.2.22.2}$$

1.2.23 Find the ratio in which the y-axis divides the line segment joining the points $(6, -4)$ and $(-2, -7)$. Also find the point of intersection.

solution :

equation of line = $3x - 8y = 50$, this line cuts the y- axis at $(0, -\frac{25}{4})$
let $\lambda : 1$ be the ratio , using section formula
$(0, -\frac{25}{4}) = \left(\frac{-2\lambda+6}{\lambda+1}, \frac{-7\lambda-4}{\lambda+1}\right)$
$\lambda = 3$, so ratio is 3:1 and intersection point is $(0, -\frac{25}{4})$

1.2.24 Use Euclid Division Lemma to show that the square of any positive integer is either of the form $3q$ or $3q + 1$ for some integer q.

Solution:

any number can be represented by 3n ,3n+1 or 3n+2 .
$p = (3n)^2 = 9n^2 = 3q$, where q is some integer
$r = (3n + 1)^2 = 9n^2 + 6n + 1 = 3q + 1$, where q is some integer
$r = (3n + 2)^2 = 9n^2 + 12n + 4 = 3(3n^2 + 4n + 1) + 1 = 3q + 1$, where q is some integer
so square of any integer can be shown in the form of 3q,3q+1, whereq is some integer

1.2.25 Sum of the areas of two squares is 544 m^2. If the difference of their perimeters is 32 m, find the sides of the two squares

Solution:

let a , b be side of each square ,sum of area of two squares =$544m^2$
$a^2 + b^2 = 544m^2$
difference of perimeter = 32m
4(a-b)=32m , a-b=8m
$a^2 - 8a - 240 = 0 \implies a = 20; b = 12$

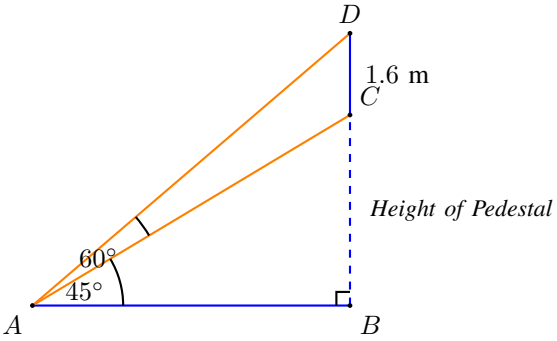
1.2.26 A motorboat whose speed is 18 km/h in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream.

Solution:
let v be speed of stream , d=24km
$t = \frac{d}{u}$
Formula :

$$\frac{d}{18-v} = 1 + \frac{d}{18+v} \tag{1.2.26.1}$$

$$48v = 324 - v^2 \implies v = 6km/h$$

1.2.27 A statue 1.6 m tall, stands on the top of a pedestal. From a point on the ground, the angle of elevation of the top of the statue is $60°$ and from the same point the angle of elevation of the top of the pedestal is $45°$. Find the height of the pedestal. (Use $\sqrt{3} = 1 \cdot 73$)
solution :



let distance from point to lower point of pedestal be x ;
$tan45° = \frac{BC}{AB} = BC = AB = x$
$tan60° = \frac{BD}{AB} = \sqrt{3} \implies \frac{x+1.6}{x} = \sqrt{3}$
height of pedestal =x=$\frac{1.6}{\sqrt{3}-1} = 0.8(\sqrt{3}+1)$

1.2.28 Show that the points $(7, 10)$, $(-2, 5)$ and $(3, -4)$ are vertices of an isosceles right triangle.
Solution:
Let the points be $A(7, 10)$, $B(-2, 5)$, and $C(3, -4)$.
Distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$$AB^2 = (-2-7)^2 + (5-10)^2 = (-9)^2 + (-5)^2 = 81 + 25 = 106$$
$$BC^2 = (3-(-2))^2 + (-4-5)^2 = (5)^2 + (-9)^2 = 25 + 81 = 106$$
$$AC^2 = (3-7)^2 + (-4-10)^2 = (-4)^2 + (-14)^2 = 16 + 196 = 212$$

Since $AB^2 = BC^2 = 106$, we have $AB = BC = \sqrt{106}$.
Because two sides are equal, ΔABC is an isosceles triangle.
3. Verify Right-Angled Property Using the converse of Pythagoras' theorem:

$$AB^2 + BC^2 = 106 + 106 = 212 = AC^2$$

Since $AB^2 + BC^2 = AC^2$, ΔABC is a right-angled triangle at B.
Thus, the given points are the vertices of an isosceles right triangle.

1.2.29 Obtain other zeroes of the polynomial $p(x) = 2x^4 - x^3 - 11x^2 + 5x + 5$ if two of its zeroes are $\sqrt{5}$ and $-\sqrt{5}$ [Q.38]

Solution:

let x,y be the other two solutions ;
sum of solutions = $-\frac{b}{a} = \frac{1}{2}$
$x + y = \frac{1}{2}$ and product of solutions = $\frac{e}{a} \implies -5xy = \frac{5}{2}$
$xy = -\frac{1}{2} \implies$ x,y are the roots of $2x^2 - x - 1 = 0$
1 and $-\frac{1}{2}$ are the other two roots of the equation $2x^4 - x^3 - 11x^2 + 5x + 5$

codes/10/q38.py (1.2.29.1)

1.2.30 What minimum must be added to $2x^3 - 3x^2 + 6x + 7$ so that the resulting polynomial will be divisible by $x^2 - 4x + 8$?

solution :

$$P(x) = Q(x)Y(x) + R(x) \implies p(x) = 2x^3 - 3x^2 + 6x + 7 \tag{1.2.30.1}$$

$Y(x) = x^2 - 4x + 8 \implies R(x) = 10x - 33$
$P(x) - R(x) = 2x^3 - 3x^2 - 4x + 40$ is divisible by $x^2 - 4x + 8$

1.2.31 The distribution given below shows the number of wickets taken by bowlers in one-day cricket matches. Find the mean and the median of the number of wickets taken.

number of wickets	20-60	60-100	100-140	140-180	180-220	220-260
number of persons	7	5	16	12	2	3

Solution:

Class Interval	**Frequency** (f_i)	**Mid-point** (x_i)	$u_i = \frac{x_i - A}{h}$	$f_i u_i$	**Cumu**
20–60	7	40	-2	-14	
60–100	5	80	-1	-5	
100–140	16	120	0	0	
140–180	12	160	1	12	
180–220	2	200	2	4	
220–260	3	240	3	9	
Total	$N = \sum f_i = 45$			$\sum f_i u_i = 6$	

Using the Step-Deviation Formula:

$$\text{Mean } (\bar{x}) = A + \left(\frac{\sum f_i u_i}{N}\right) \times h$$

Substitute $A = 120$, $\sum f_i u_i = 6$, $N = 45$, and $h = 40$:

$$\bar{x} = 120 + \left(\frac{6}{45}\right) \times 40$$

$$\bar{x} = 120 + \left(\frac{2}{15}\right) \times 40 = 120 + \frac{16}{3} = 120 + 5.33 = 125.33$$

a) Find $\dfrac{N}{2}$:

$$\frac{N}{2} = \frac{45}{2} = 22.5$$

b) The cumulative frequency just greater than 22.5 is 28, which corresponds to the median class **100–140**.

c) Values for the formula:

- Lower limit of median class (l) = 100
- Cumulative frequency of preceding class (cf) = 12
- Frequency of median class (f) = 16
- Class width (h) = 40

Apply the Median formula:

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h \tag{1.2.31.1}$$

$$\text{Median} = 100 + \left(\frac{22.5 - 12}{16}\right) \times 40 \tag{1.2.31.2}$$

$$\text{Median} = 100 + \left(\frac{10.5}{16}\right) \times 40 = 100 + \frac{10.5 \times 5}{2} = 100 + \frac{52.5}{2} = 100 + 26.25 = 126.25$$

Mean: 125.33 wickets

Median: 126.25 wickets

1.2.32 In a cylindrical vessel of radius 10 cm, containing some water, 9000 small spherical balls are dropped which are completely immersed in water which raises the water level. If each spherical ball is of radius $0 \cdot 5$ cm, then find the rise in the level of water in the vessel.

Solution:

Radius of the cylindrical vessel (R) = 10 cm

Number of spherical balls (n) = 9000

Radius of each spherical ball (r) = 0.5 cm = $\frac{1}{2}$ cm

Let the rise in the water level be h cm.

Formulae

a) Volume of a sphere = $\frac{4}{3}\pi r^3$

b) Volume of a cylinder = $\pi R^2 h$

The volume of water raised in the cylindrical vessel is equal to the total volume of 9000 spherical balls dropped into it: Volume of raised water in cylinder = 9000 × Volume of one spherical ball

Substitute the respective geometric formulae: $\pi R^2 h = 9000 \times \left(\frac{4}{3}\pi r^3\right)$

Cancel π from both sides:

$$R^2 h = 9000 \times \frac{4}{3} \times r^3$$

Substitute $R = 10$ and $r = 0.5$:

$$(10)^2 \times h = 9000 \times \frac{4}{3} \times (0.5)^3$$

Simplify the terms:

$$100 \times h = 3000 \times 4 \times 0.125$$

$$100 \times h = 12000 \times 0.125$$

$$100 \times h = 1500$$

Solve for h:

$$h = \frac{1500}{100} = 15 \text{ cm}$$

Answer

The rise in the level of water in the vessel is 15 **cm**.

1.2.33 If a line is drawn parallel to one side of a triangle to intersect the other two sides at distinct points, prove that the other two sides are divided in the same ratio

Solution:

1. Given: In ΔABC, a line DE is drawn parallel to side BC ($DE \parallel BC$), intersecting AB at D and AC at E.

2. To Prove:

$$\frac{AD}{DB} = \frac{AE}{EC} \tag{1.2.33.1}$$

Join B to E and C to D.

Draw $DM \perp AC$ (perpendicular from D to AC).

Draw $EN \perp AB$ (perpendicular from E to AB)

3. Step-by-Step Derivation: 1. Calculate the area of ΔADE considering AD as the base:

$$\text{Area}(\Delta ADE) = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times AD \times EN \tag{1.2.33.2}$$

2. Calculate the area of ΔBDE considering BD as the base:

$$\text{Area}(\Delta BDE) = \frac{1}{2} \times BD \times EN \tag{1.2.33.3}$$

3. Divide :

$$\frac{\text{Area}(\Delta ADE)}{\text{Area}(\Delta BDE)} = \frac{\frac{1}{2} \times AD \times EN}{\frac{1}{2} \times BD \times EN} = \frac{AD}{BD} \tag{1.2.33.4}$$

4. Similarly, calculate the area of ΔADE with base AE and height DM, and ΔDEC with base EC and height DM:

$$\frac{\text{Area}(\Delta ADE)}{\text{Area}(\Delta DEC)} = \frac{\frac{1}{2} \times AE \times DM}{\frac{1}{2} \times EC \times DM} = \frac{AE}{EC} \tag{1.2.33.5}$$

5. Relate the areas of ΔBDE and ΔDEC:

Both ΔBDE and ΔDEC lie on the same base DE and between the same parallel lines DE and BC.

Since triangles on the same base and between the same parallels are equal in area:

$$\text{Area}(\Delta BDE) = \text{Area}(\Delta DEC) \quad \text{— (Equation 3)} \tag{1.2.33.6}$$

- Comparing Equations (1), (2), and (3), their left-hand sides are equal because they have identical numerators (Area(ΔADE)) and equal denominators (Area(ΔBDE) = Area(ΔDEC)).
- Therefore, their right-hand sides must be equal:

$$\frac{AD}{DB} = \frac{AE}{EC}$$